TOPIC 2: ATOMIC STRUCTURE AND THE PERIODIC TABLE

GRADE 11

SEMESTER 1/2021

Students will be assessed on their ability to:		***Checklist*** ✓
2.1	know the structure of an atom in terms of electrons, protons and neutrons	☐
2.2	know the relative mass and charge of protons, neutrons and electrons	☐
2.3	know what is meant by the terms 'atomic (proton) number' and 'mass number'	☐
2.4	be able to use the atomic number and the mass number to determine the number of each type of subatomic particle in an atom or ion	☐
2.5	understand the term 'isotope'	
2.6	understand the basic principles of a mass spectrometer and be able to analyse and interpret mass spectra to:	
	i deduce the isotopic composition of a sample of an element	☐
	ii calculate the relative atomic mass of an element from relative abundances of isotopes and vice versa	☐
	iii determine the relative molecular mass of a molecule, and hence identify molecules in a sample	☐
	iv understand that ions in a mass spectrometer may have a 2+ charge	☐
2.7	be able to predict mass spectra, including relative peak heights, for diatomic molecules, including chlorine, given the isotopic abundances	☐
2.8	be able to define first, second and third ionisation energies and understand that all ionisation energies are endothermic	☐
2.9	know that an orbital is a region within an atom that can hold up to two electrons with opposite spins	☐
2.10	understand how ionisation energies are influenced by the number of protons in the nucleus, the electron shielding and the sub-shell from which the electron is removed	☐
2.11	know that ideas about electronic structure developed from:	
	i an understanding that successive ionisation energies provide evidence for the existence of quantum shells and the group to which the element belongs	☐
	ii an understanding that the first ionisation energy of successive elements provides evidence for electron sub-shells	☐
2.12	be able to describe the shapes of *s* and *p* orbitals	☐
2.13	know that orbitals in sub-shells:	
	i each take a single electron before pairing up	☐
	ii pair up with two electrons of opposite spin	☐
2.14	be able to predict the electronic configuration of atoms of the elements from hydrogen to krypton inclusive and their ions, using *s, p, d* notation and electron-in-boxes notation	☐
2.15	understand that electronic configuration determines the chemical properties of an element	☐

2.16 know that the Periodic Table is divided into blocks, such as *s, p* and *d,* and know the number of electrons that can occupy *s, p* and *d* sub-shells in the first four quantum shells ☐

2.17 be able to represent data, in a graphical form (including the use of logarithms of first ionisation energies on a graph) for elements 1 to 36 and hence explain the meaning of the term 'periodic property' ☐

2.18 be able to explain: ☐

i the trends in melting and boiling temperatures of the elements of Periods 2 and 3 of the Periodic Table in terms of the structure of the element and the bonding between its atoms or molecules ☐

ii the general increase and the specific trends in ionisation energy of the elements across Periods 2 and 3 of the Periodic Table ☐

iii the decrease in first ionisation energy down a group ☐

2A – 1 STRUCTURE OF ATOMS AND ISOTOPES

SPECIFICATION OBJECTIVE 2.1, 2.2, 2.3, 2.4, 2.5

Our current understanding of the structure of atoms is influenced by the theories put forward by scientists such as J.J Thomson (discovered electrons in 1897), Ernest Rutherford (discovered proton in 1917) and James Chadwick (discovered the neutron in 1932).

STRUCTURE OF AN ATOM

- An atom is the smallest, electrically neutral particle of an element that can take part in a chemical change.
- An atom comprises of subatomic particles **protons**, **neutrons** and **electrons**.
- Protons and neutrons, together known as **nucleons**, are found in the nucleus which is the center of the atom.
- Electrons exist in energy levels surrounding the nuclei. The energy levels are called **quantum shells**.
- An atom gets its chemical properties from the number of electrons and arrangement of electrons in the atom.

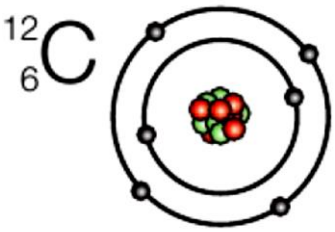
Table A. The structure of the atom in terms of the three sub atomic particles

Particle	Symbol	Charge/C	Relative charge	Mass/g	Relative mass
Proton	p	$+1.60 \times 10^{-19}$	+1	1.67×10^{-24}	1
Neutron	n	0	0	1.67×10^{-24}	1
Electron	e^-	-1.60×10^{-19}	-1	9.11×10^{-28}	1 / (1840)

- The mass of electrons are negligible, hence the mass of the atoms are determined by the mass of nucleons.

ATOMIC NUMBER, MASS NUMBER AND ISOTOPES

The figure below shows the atomic number, mass number and isotopes of carbon



Carbon atoms

Mass number → $^{12}_{6}C$, $^{13}_{6}C$

Atomic number →

Atomic number = number of protons (= number of electrons)

Mass number = number of protons + number of neutrons

ISOTOPES

- Isotopes are atoms of the same element having the same number of protons but different number of neutrons.
- Example: Carbon has three isotopes, ^{12}C, ^{13}C and ^{14}C. They all have the same number of protons but different number of neutrons.

	Protons	Neutrons	Mass number
Carbon – 12	6	6	12
Carbon – 13	6	7	13
Carbon – 14	6	8	14

- Isotopes always have the **same chemical properties** because the **number of electrons is the same**.
- They have slightly different physical properties as they have different number of neutrons. ***Example:*** Isotopes with fewer neutrons will have lower masses, lower densities, faster rate of diffusion, lower melting and boiling points.
- Relative atomic masses measured by chemical methods rarely produce whole numbers. This is because isotopes of the same element exist in different proportions in nature.

EXERCISE 1

Complete the following table

	Protons	Neutrons	Electrons	Charge	Atomic No.	Mass No.	Symbol
A	19	21	19				
B	20			Neutral		40	
C				+	11	23	
D	6	6		Neutral			
E	92			Neutral		235	
F	6		6			13	
G		16		2-	16		
H							$^{27}Al^{3+}$

CHECKPOINT 1

Do the check point questions on page 43 from text book.

2A - 2 MASS SPECTROMETRY AND RELATIVE MASSES OF ATOMS, ISOTOPES AND MOLECULES

SPECIFICATION OBJECTIVE 2.6 & 2.7

RELATIVE MASSES OF ELEMENTS OR COMPOUNDS

- Measuring the mass of an atom or compound using a balance would be very difficult as they are very small.
- Hence, to measure the mass of an atom or a compound the ***atomic mass unit*** (amu) is used as a standard reference.
- The relative atomic mass does not have any units. They are measured compare to amu.
    - ***Atomic mass unit*** is the mass of a carbon-12 atom divided by 12. This is used as the standard reference scale to measure mass of atoms and molecules.
    - ***Relative atomic mass*** of an element is the weighted average mass of an atom of that element divided by $1/12^{th}$ the mass of a carbon-12 atom.
    - ***Relative molecular mass*** of a molecule or compound is the sum of the relative atomic masses of all the atoms in the chemical formula.
    - ***Relative isotopic mass*** is the mass of an atom of that isotope divided by $1/12^{th}$ the mass of a carbon-12 atom.

USING DATA OBTAINED FROM A MASS SPECTROMETER

WHAT IS A MASS SPECTROMETER?

- A mass spectrometer is an instrument that was invented by F.W. Aston in 1919. It can be used to find the relative masses of isotopes of atoms, molecules or fragments of molecules. This gave chemists an accurate method of comparing the relative masses of atoms and molecules. And to find the relative abundance of the isotopes in a sample of an element.
- The mass spectrometer consists of an ion source, an analyser and a detector.
- Inside the spectrometer there is a high vacuum which allows ionized atoms and molecules from the chemical being tested to be studied without interference from atoms and molecules in the air.
- Mass spectrometry can be used for:
    - Carbon dating – to measure the amount of C-14 in a substance and hence to determine the age of the substance
    - Pharmaceutical industry - to identify the new compounds in possibly new drugs and to identify how long they would stay in the body
    - Testing athletes urine for drugs or banned substances such as anabolic steroids
    - Studying the composition of the surface of Mars and to look for molecules that might suggest that life existed on the planet by sending probes to Mars.

HOW DOES A MASS SPECTROMETER WORK?

There are five main processes occurring in a mass spectrometer. They are:

1. Vapourisation: the sample is turned into gas or vapourised using an electrical heater.

$$X(s) \rightarrow X(g)$$

$$X(l) \rightarrow X(g)$$

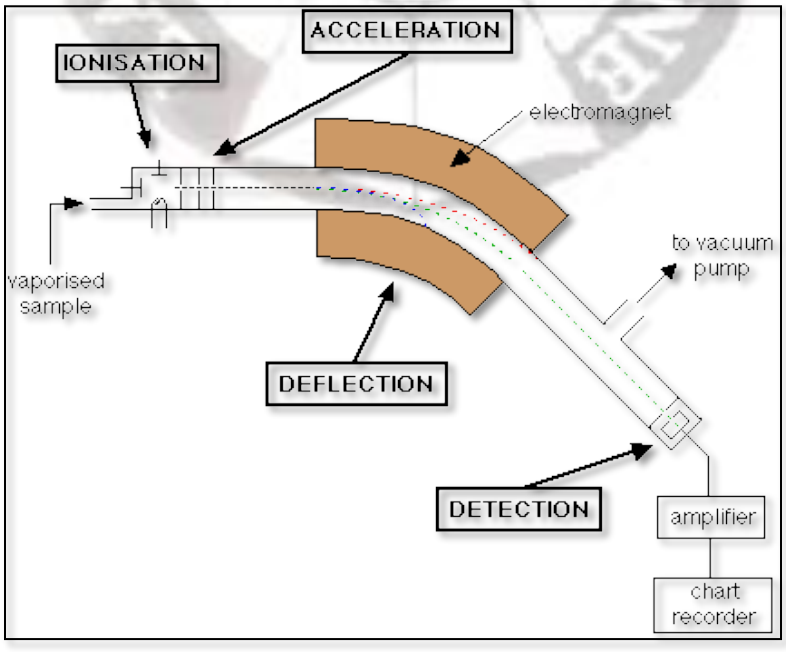
2. Ionization: gas particles are bombarded with a stream of high energy electrons from an electron gun, to ionize them. Electrons are knocked off the particles, producing uni-positively charged ions

$$X(g) \rightarrow X^{+}(g) + e^{-}$$

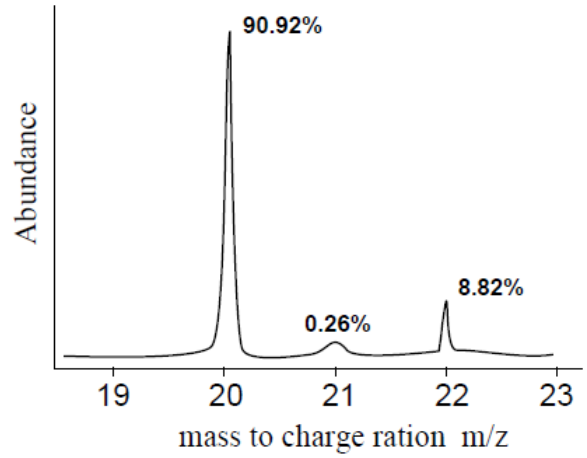
3. Acceleration: the positive ions are accelerated by an electric field towards the magnetic field, which provides the energy for the particles to accelerate.

4. Deflection: The positive ions' path is altered with a magnetic field. For a given magnetic field, only ions with a particular mass/charge ratio will be detected. Lighter ions have less momentum and are deflected more than heavier ions. Ions with a large m/z ratio deflected less than ions with a smaller m/z ratio.

5. Detection: the ions are detected by the detector, which is linked to an amplifier and then to a recorder. This converts the current to a peak which is shown in a mass spectrum.



Mass spectrum



- An early application was the demonstration by Aston, (Nobel Prize, 1922), that naturally occurring neon consisted of three isotopes;

  ^{20}Ne, ^{21}Ne and ^{22}Ne

- position of peaks give atomic mass
- peak intensity gives relative abundance
- The tallest peak represents the most abundant isotope.

DETERMINING RELATIVE ISOTOPIC AND ATOMIC MASSES

The mass spectrum could be used to calculate the relative atomic mass of an element.

WORKED EXAMPLE 1

Calculate the average relative atomic mass of neon using the above information. Out of every 100 atoms 90.92 are ^{20}Ne, 0.26 are ^{21}Ne and 8.82 are ^{22}Ne.

WORKED EXAMPLE 2

What is the relative atomic mass of chlorine, if it exists in two isotopic forms shown in the table below?

Isotope	Relative isotopic mass	Relative abundance
$^{35}Cl_{17}$	35	75
$^{37}Cl_{17}$	37	25

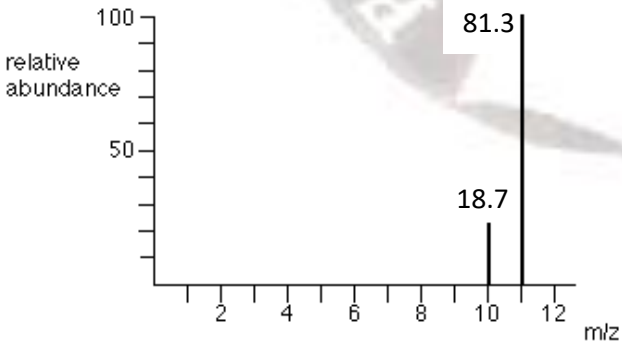
EXERCISE 2

Question 1: Calculate the relative atomic mass of sulfur from the following isotopic percentages

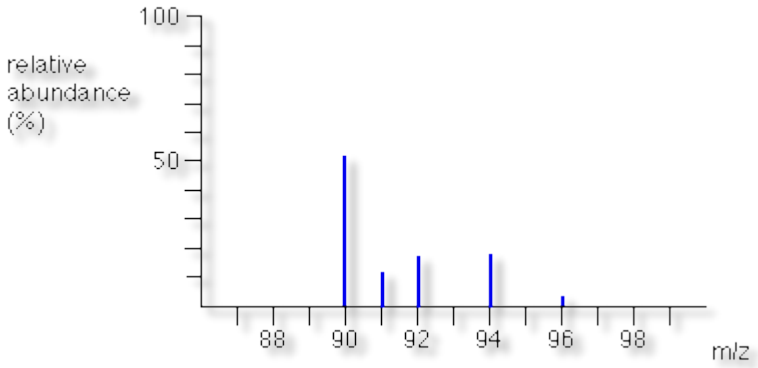
^{32}S 95% ^{33}S 1% ^{34}S 4%

Question 2: The neon element has three isotopes. They are 90.92% of ^{20}Ne, 0.26% of ^{21}Ne and 8.82% of ^{22}Ne. Calculate relative atomic mass.

Question 3: Calculate the relative atomic mass of boron using the mass spectrum.



Question 4: Calculate the relative atomic mass of Zirconium using the mass spectrum.



zirconium-90: 51.5

zirconium-91: 11.2

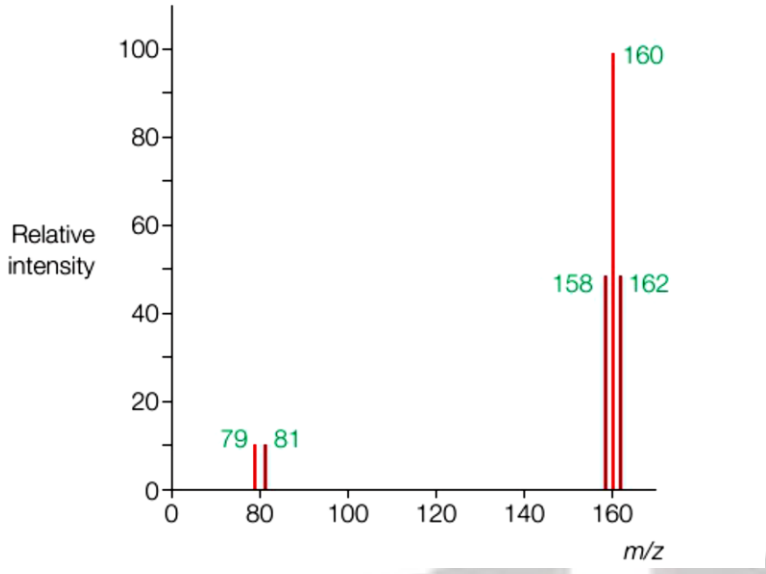
zirconium-92: 17.1

zirconium-94: 17.4

zirconium-96: 2.8

Question 5: Naturally occurring potassium consists of potassium-39 and potassium-41. Calculate the percentage of each isotope present if the average is 39.1.

Question 6: The mass spectrum of bromine vapour is shown below



(a) What are the relative isotopic masses of the two isotopes present in bromine?

(b) Identify the particles responsible for the peaks at m/z 158, 160 and 162

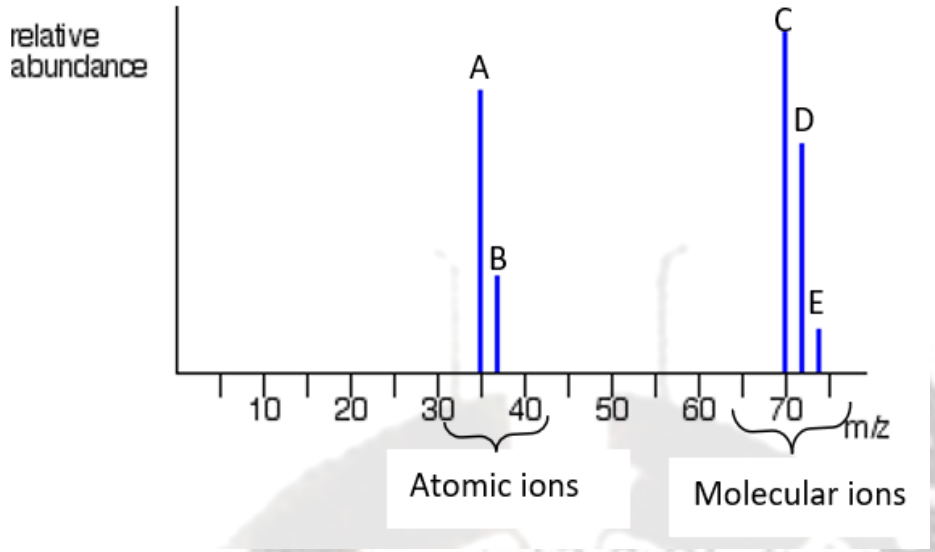
(c) Deduce the relative abundance of the two isotopes and explain the relative heights of the three peaks at m/z 158, 160 and 162

CHECKPOINT 2

Do the check point questions on page 47 from text book.

DETERMINING RELATIVE MOLECULAR MASS (M_r) OF DIATOMIC MOLECULES

If an element exists as diatomic molecules, like chlorine, then the spectrum would contain peaks for the separate atomic ions as well as for the molecular ions.



There are two naturally occurring isotopes of chlorine. Hence 3 different combinations of Cl_2 molecules can be obtained. This is shown by the peaks C, D and E.

If diatomic substances are analysed by mass spectrometry, you can obtain the relative molecular mass of the element or compound by observing the peaks with the largest m/z ratios (assuming a value of z=1).

Molecular ions are formed by the removal of an electron from the whole molecule. Atomic ions are formed when the Cl—Cl bond is broken by a high energy beam of electron.

Peak	Peak height ratio	m/z ratio	Species
A	3	35	
B	1	37	
C	9	70	
D	6	72	
E	1	74	

Explanation of peak height ratio

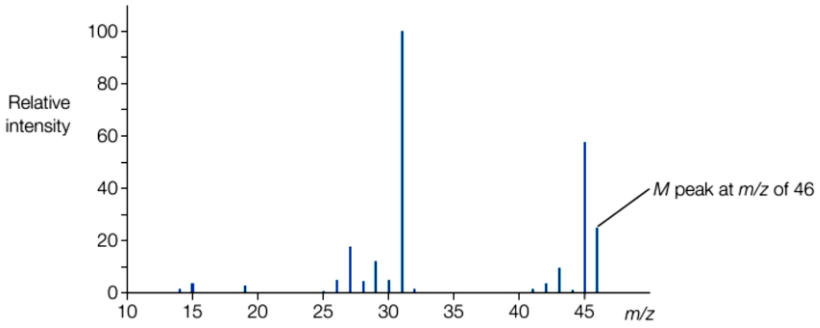
- There are two isotopes of chlorine, ^{35}Cl which is 75% abundant and ^{37}Cl which is 25% abundant.

Isotope	^{35}Cl	^{37}Cl	
Abundance	75%	25%	= 100%
Chance of getting each isotope	$\frac{75}{100}$	$\frac{25}{100}$	
Simplifying the ratio	$\frac{3}{4}$	$\frac{1}{4}$	

- There is 3 in 4 chance of selecting a ^{35}Cl from a sample of chlorine atom. This means total chance of two ^{35}Cl atoms combining together is ¾ X ¾ = $\frac{9}{16}$
- There is 1 in 4 chance of selecting a ^{37}Cl atom. This means that the chance of ^{37}Cl combining with a ^{35}Cl atom is ¼ X ¾ = $\frac{3}{16}$
- The chance of a ^{35}Cl atom combining with a ^{37}Cl atom is also $\frac{3}{16}$
- This means that the total chance of ^{37}Cl and ^{35}Cl combining together in any order is 2 X $\frac{3}{16}$ = $\frac{6}{16}$
- The chance of two ^{37}Cl atoms combining together is ¼ X ¼ = $\frac{1}{16}$
- $\frac{9}{16}$: $\frac{6}{16}$: $\frac{1}{16}$ in whole number ratio is 9 : 6 : 1

DETERMINING RELATIVE MOLECULAR MASS OF A POLYATOMIC MOLECULE

Mass spectrum of ethanol (C_2H_5OH)



*The peak with the highest m/z ratio is the **molecular ion peak** which gives the RMM of the compound.*

RMM for ethanol = 46

2A – 3 ATOMIC ORBITALS AND ELECTRONIC CONFIGURATION

SPECIFICATION OBJECTIVE 2.9, 2.12, 2.13, 2.14, 2.15

QUANTUM SHELLS

Atoms can be identified by its mass of their nucleons; however, it is their electronic structure which determines the behavior and chemical properties. Max Planck first presented the quantum theory in 1900. We can use this theory to describe the arrangement of electrons around the nuclei.

According to this theory electrons exist around the nucleus in certain regions of the atom called shells or energy levels. These well-defined energy levels are called quantum shells.

Each shell is described by a number known as principle quantum number (n). The larger the value of n, the further the electron is from the nucleus.

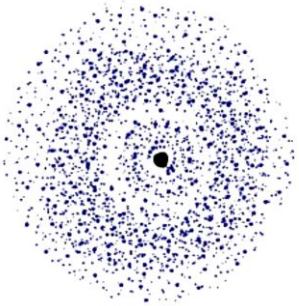
ELECTRONS IN THE FIRST FOUR QUANTUM SHELLS

- The energy levels or quantum shells (apart from the first one) are divided into sub-shells of slightly different energy levels. Each subshell is described by a letter s, p, d and f.
- The energy of each sub-shell increases from s to p to d to f.

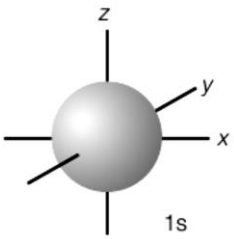
Quantum shell or Energy level	Sub -shell
1	1s
2	2s 2p
3	3s 3p 3d
4	4s 4p 4d 4f

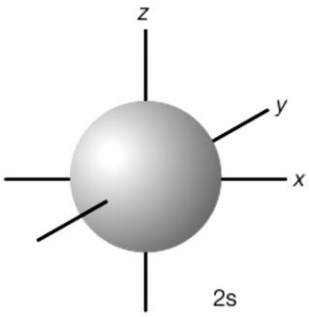
- Each sub-shell is made up of orbitals.
  - Orbital is the region where the electrons are most likely to be found.
    - 3-dimensional statistical maps showing the likeliest places to find electrons
    - come in different shapes and sizes
  - The orbitals in different sub – levels have different shapes

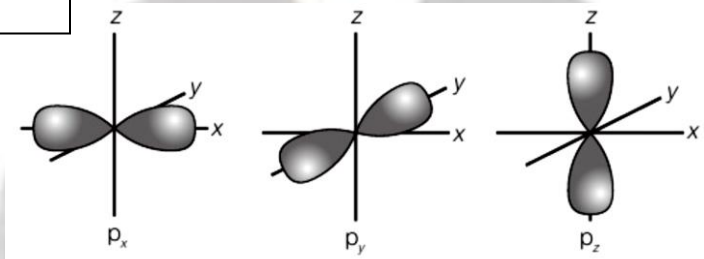
s- orbital



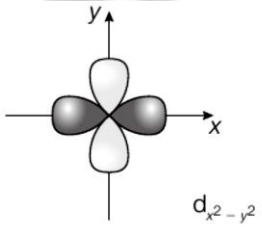
A cross section through an s - orbital

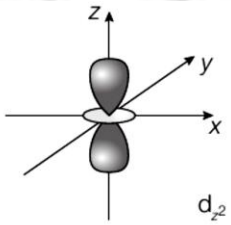


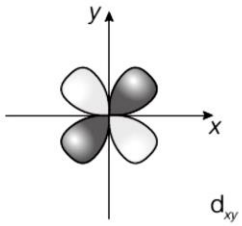


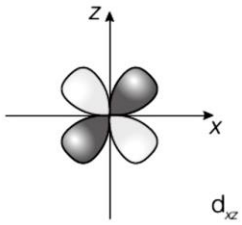


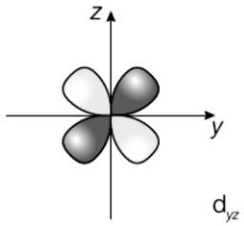
d- orbital











Each orbital can accommodate a **maximum of two electrons** in which one electron spins clockwise and the other spins anti – clockwise. ↑↓

Table showing the number of electrons that fit in each type of sub – shell

Sub – shell	Number of orbitals	Total number of electrons
s	1	(1 x 2) 2
p	3	(3 x 2) 6
d	5	(5 x 2) 10
f	7	(7 x 2) 14

Table showing the sub – shells and electrons in the first four energy levels

Shell	Principle Quantum number (n)	Sub – shell	Total number of electrons
1st	1	1s	2
2nd	2	2s 2p	8
3rd	3	3s 3p 3d	18
4th	4	4s 4p 4d 4f	32

ELECTRONIC CONFIGURATION

Electrons are added to the sub shells according to the **Aufbau Principle.** The following rules must be applied when writing electronic configuration.

Aufbau principle states that;

- Electrons fill up the lowest energy orbital first.

Hund's rule states that;

- Electrons fill orbitals singly before they start pairing up according to Hund's rule

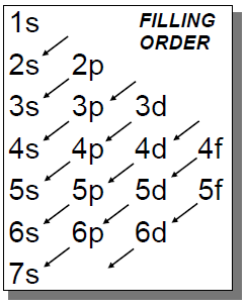
The Pauli Exclusion Principle states that;

- two electrons cannot occupy the same orbital unless they have opposite spin. This is due to the repulsion between electron pairs.

- The sub shells which are **half filled** or **completely filled** with electrons are **more stable**.

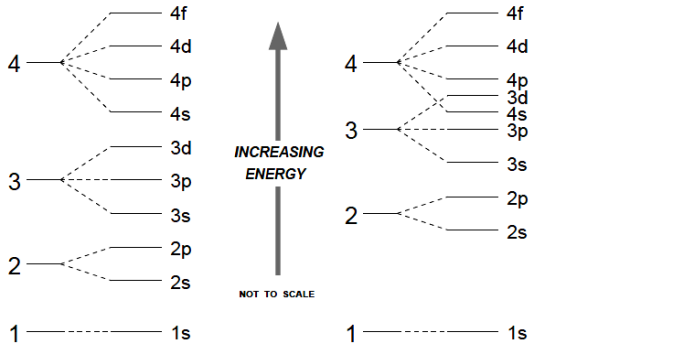
Orbitals are not filled in numerical order.

For example, 4s is filled before 3d. This is because the 4s orbital is at a lower energy level than a 3d orbital.



Hence the order is

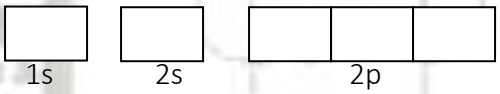
1s 2s 2p 3s 3p 4s 3d 4p……



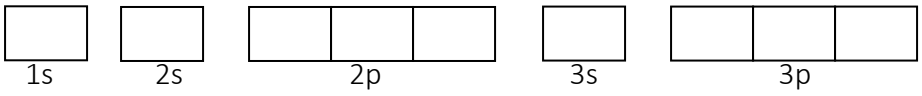
The diagram helps explain why the 4s orbitals are filled before the 3d orbitals

When writing electronic configuration, the electronic structure of an atom can be represented in **two** ways:

1. **s p d f notation**
2. **electron box notation**

EXAMPLE 1
Write down the electronic configuration of $_{6}C$ using spdf notation and box notation **s p d f notation** : **Box notation** : 

A shortened version of electronic structure can be used to represent electronic configuration. That is, using **noble gas electronic configuration**

EXAMPLE 2
Write down the electronic configuration of $_{16}S$ using spdf notation and box notation **s p d f notation** : **Box notation** :    **Noble gas electronic configuration:**

EXERCISE 3

Write electronic configuration for elements from hydrogen, H to Krypton, Kr using both s p d f notation and box notation.

Atomic number	Element	s p d notation and box notation
1	H	
2	He	
3	Li	
4	Be	
5	B	
6	C	
7	N	
8	O	
9	F	
10	Ne	

11	**Na**	
12	**Mg**	
13	**Al**	
14	**Si**	
15	**P**	
16	**S**	
17	**Cl**	
18	**Ar**	
19	**K**	
20	**Ca**	

21	**Sc**	
22	**Ti**	
23	**V**	
24	**Cr**	
25	**Mn**	
26	**Fe**	
27	**Co**	
28	**Ni**	
29	**Cu**	
30	**Zn**	

31	**Ga**	
32	**Ge**	
33	**As**	
34	**Se**	
35	**Br**	
36	**Kr**	

The electronic structure of **Cr and Cu are exceptional**. In both cases the outer subshells are either completely full or half full. This is achieved by donating one of their 4s electrons to the 3d sub shell as this makes the d- sub shell to be half filled or completely filled.

Chromium is $1s^2 2s^2 2p^6 3s^2 3p^6 \mathbf{3d^5 4s^1}$

Not $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$

Copper is $1s^2 2s^2 2p^6 3s^2 3p^6 \mathbf{3d^{10} 4s^1}$

Not $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^2$

Completely filled or half filled-sub shell is a more stable electronic configuration.

CHECKPOINT 3

Do the check point questions on page 51 from text book.

ELECTRONIC CONFIGURATION OF IONS

When an atom loses or gains an electron to form ions, the electronic structure changes.

Cations (positive ions) – formed by loss of electrons

Anions (negative ions) – formed by gain of electrons

EXAMPLE 1
Write down the electronic configuration of sodium atom and sodium ion using spdf notation Sodium atom, $_{11}Na$ Sodium ion, Na^+

EXAMPLE 2
Write down the electronic configuration of chlorine atom and chloride ion using spdf notation Chlorine atom, $_{17}Cl$ chloride ion, Cl^-

EXERCISE 4

Give the full electronic structure of the following ions

a. Mg^{2+} :

b. Ca^{2+} :

c. Al^{3+} :

d. O^{2-} :

e. Br^- :

f. P^{3-} :

2A – 3 IONISATION ENERGIES

SPECIFICATION OBJECTIVE 2.9, 2.12, 2.13, 2.14, 2.15

IONIZATION ENERGY

- It is a measure of the energy required to remove electrons from an atom.
- There are as many ionization energies as there are electrons in the atom.
- The value of ionization energy depends on
  - the distance of the electron from the nucleus, and
  - the effective nuclear charge (not the nuclear charge) of the atom.

Nuclear Charge (**NC**) :The actual charge (relative) due to the protons in the nucleus

Effective Nuclear Charge (**ENC**) :The effectiveness of nuclear charge after passing through filled shells

A simple way to compare effective nuclear charges is to knock off a **+** for every electron in a filled inner level.

Species	*Protons*	*Electron config.*	*NC*	*ENC*
H	1	1	1+	1+
He	2	2	2+	2+
Li	3	2,1	3+	1+
Be	4	2,2	4+	2+
B	5	2,3	5+	3+
Ne	10	2,8	10+	8+
Na	11	2,8,1	11+	1+
K	19	2,8,8,1	19+	1+

FIRST IONIZATION ENERGY

Definition :

The first ionisation energy is the energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous uni-positive ions.

This is represented by the equation:

Eg: $Na(g) \rightarrow Na^+(g) + e^-$

$Mg(g) \rightarrow Mg^+(g) + e^-$

Always gaseous

1	H	1310	15	P	1060
2	He	2370	16	S	1000
3	Li	519	17	Cl	1260
4	Be	900	18	Ar	1520
5	B	799	19	K	418
6	C	1090	20	Ca	590
7	N	1400	21	Sc	632
8	O	1310	22	Ti	661
9	F	1680	23	V	648
10	Ne	2080	24	Cr	653
11	Na	494	25	Mn	716
12	Mg	736	26	Fe	762
13	Al	577	27	Co	757
14	Si	786	28	Ni	736

Table: First Ionisation Energies / kJ mol^{-1}

Its value gives an idea of how strongly the nucleus pulls on the electron being removed. The stronger the pull, the more energy needed to remove the electron.

SECOND IONIZATION ENERGY

Definition :

The second ionization energy is the energy required to remove 1 mole of electrons from 1 mole of the gaseous uni-positive ions to form one mole of gaseous di-positive ions.

Eg: $Na^+(g) \rightarrow Na^{2+}(g) + e^-$

$Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$

THIRD IONIZATION ENERGY

Definition :

The third ionization energy is the energy required to remove 1 mole of electrons from 1 mole of the gaseous di-positive ions to form one mole of gaseous tri-positive ions.

Eg: $Na^{2+}(g) \rightarrow Na^{3+}(g) + e^-$

$Mg^{2+}(g) \rightarrow Mg^{3+}(g) + e^-$

SUCCESSIVE IONIZATION ENERGIES OF AN ELEMENT

The patterns in successive ionisation energies for an element give us important information about the electronic structure for that element.

When successive ionization energies of an element are listed, there are steady increases and big jumps occuring at defined places. This is one piece of evidence for the existence of quantum shells.

- A sodium atom has 11 electrons. Its successive ionization energies are shown in the table below.
- The places where the ionization energy has jumped significantly are underlined in the table.

1st	2nd	3rd	4th	5th	6th
496	4563	6913	9544	13352	16611

7th	8th	9th	10th	11th
20115	25491	28934	141367	159079

Successive ionization energies of sodium in $kJmol^{-1}$

When a graph is plotted between the log of ionization energy values of an element against the number of electrons removed, the trends in successive ionization energies can be seen. There are two types of increases that can be observed. They are:

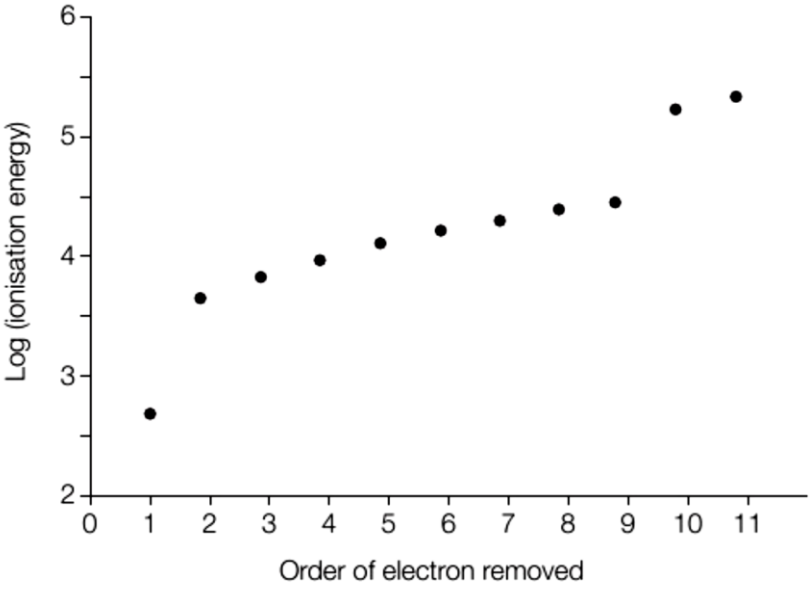
Successive ionization energies are always greater than the previous one.

- This is due to increasing attraction of the nucleus on the remaining electrons because the electrons are being removed from an increasingly positive ion.

Large increases occur when there is a change of shell

- This is due to removal of electron from an inner main shell, which is closer to the nucleus on which the nuclear attraction is greater.
- This can be used to predict the group of an unknown element. The group number would be given by *counting the number of electrons removed before the first big jump*. Electronic structure can also be found using successive ionization energies.

Graph showing successive ionization energies for Sodium



WHY DO THE SUCCESSIVE IONIZATION ENERGIES OF SODIUM INCREASES?

- The first electron to be removed is in the 3s orbital, which is the third quantum shell. So less energy is needed to remove the electron. Therefore the first ionization energy is the lowest.
- There is a large jump from 1st to the 2nd ionization energy. This is because the second electron to be removed is in the 2nd quantum shell which is lower in energy than the 3rd quantum shell
- There is steady increase in ionization energy from the 2nd to the 9th electron. This shows that the eight electrons exist within the same quantum shell. As each successive electron is removed from the same shell, the electron – electron repulsion within the shell decreases. As a result, there is steady increase in ionization energy from 2nd to 9th electron.
- The large jump from 9th to 10th electron indicates that the electron is removed from first quantum shell (1s orbital).

TRENDS IN IONIZATION ENERGIES

The amount of energy required to remove the first electron depends on the following factors:

Distance between the nucleus and the outermost electrons

- As the atoms get larger, the distance between the outer most electrons and the nucleus increases.
- The outer electrons are held less strongly so easier to remove.
- Less energy is required to remove the electron (lower ionization energy).

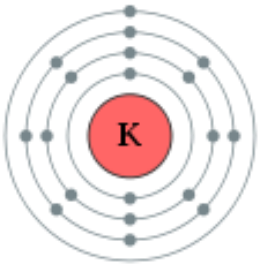
Effective nuclear force of attraction

- As the effective nuclear charge increases, force of attraction between the electrons and outer electrons increases.
- Electrons are harder to remove (higher ionization energy).

Shielding effect

- The electron in the outermost energy level experiences more inter-electron repulsion known as shielding.
- The greater the number of the inner shell electrons the greater is the shielding effect (screening effect).
- The force of attraction becomes weaker.
- Less energy is required to remove the electron (lower ionization energy).

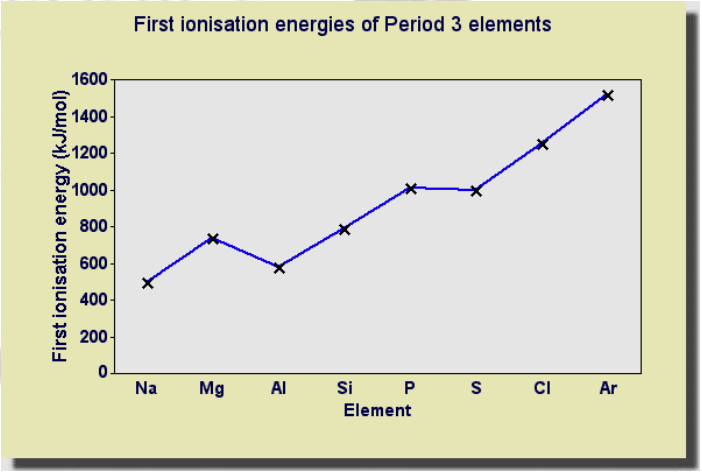
19: Potassium **2,8,8,1**



ACROSS THE PERIOD

In general the 1st ionization energy of the elements across the period **increases** gradually, because:

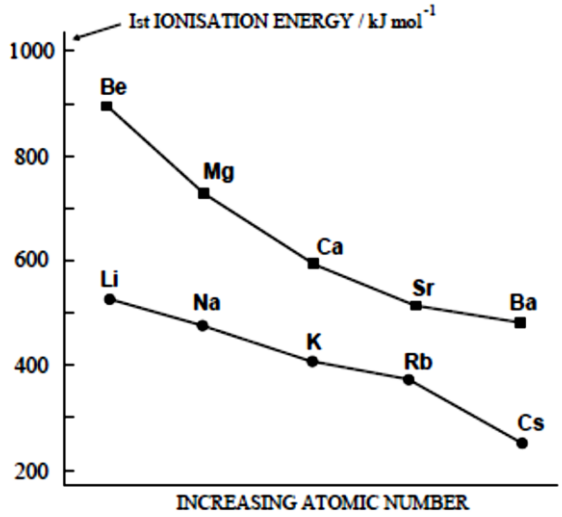
- As the atomic size decreases, distance between the nucleus and the outer shell electrons decreases
- Nuclear attraction increases
- Shielding effect remains the same
- Therefore, the attraction between the outer electrons and the nucleus increases. Hence, more energy is required to remove the electrons.



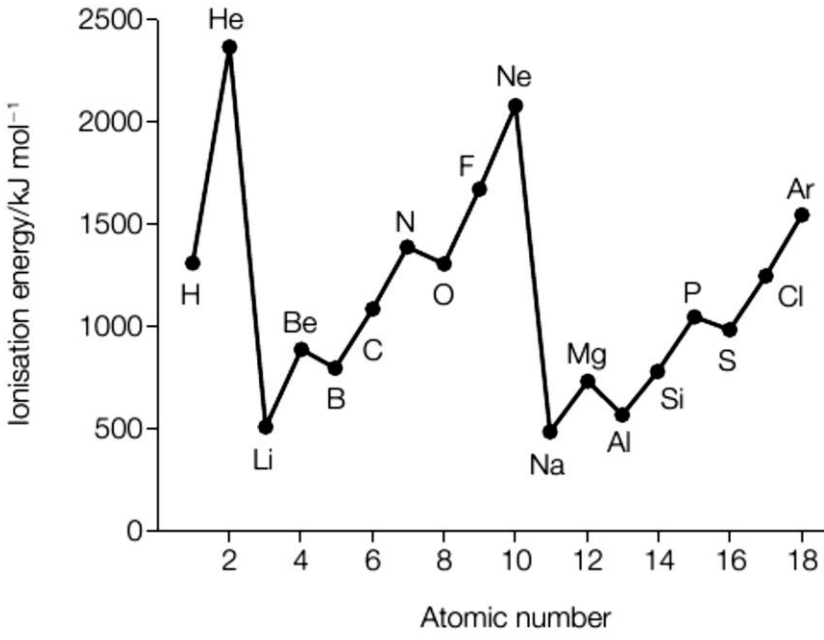
DOWN THE GROUP

Ionization energy decreases down the group. This is because :

- Down the group, the atomic size increases and the number of shells increases.
- Distance between the nucleus and the outer electrons increases.
- The outer electrons are more shielded by the electrons in the inner shells.
- The effect of the increase in nuclear charge is outweighed by the increase in shielding.
- Therefore, the outer electrons are less attracted by the nucleus. Hence, less energy is required to remove the electrons.



FIRST IONIZATION ENERGIES



First ionization energies for periods 1 - 3

First ionization energy of Helium is greater than Hydrogen [He >H]

- **Nuclear charge is greater** – one extra proton in Helium
- Extra electron has gone into the same energy level.
- Increased attraction makes the electron harder to remove.

Deviation between group 2 & 3

First ionization energy of Boron is less than Beryllium [B < Be]

Beryllium [4]: $1s^2\ 2s^2$

Boron [5]: $1s^2\ 2s^2\ 2p^1$

- Despite the increased nuclear charge, the outer electron is held less strongly
- The outermost electron of boron is in a 2p orbital, which is higher in energy than the 2s orbital of beryllium.
- Nuclear attraction is less in boron because the 2p electrons in boron experiences greater electron-electron repulsion (greater shielding) than that in beryllium.
- Less energy is required to remove an electron from boron.

First ionization energy of Aluminium is less than Magnesium [Al<Mg]

Magnesium [12]: $1s^2\ 2s^2\ 2p^6 3s^2$

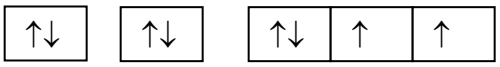
Aluminium [13]: $1s^2\ 2s^2\ 2p^6 3s^2\ 3p^1$

- Despite the increased nuclear charge, the outer electron is held less strongly
- The outermost electron of Aluminium is in a 3p orbital, which is higher in energy than the 3s orbital of Magnesium.
- Nuclear attraction is less in aluminium because the 3p electrons in aluminium experiences greater electron-electron repulsion (greater shielding) than that in magnesium.
- Less energy is required to remove an electron from aluminium.

Deviation between group 5 & 6

First ionization energy of Oxygen is less than Nitrogen [O < N]

Nitrogen [7]: $1s^2\ 2s^2\ 2p^3$ 

Oxygen[8]: $1s^2\ 2s^2\ 2p^4$ 

- Despite the increased nuclear charge the electron is easier to remove
- The 2p electron removed from oxygen is a paired electron, whereas the 2p electron from nitrogen is an unpaired electron.
- When electrons are paired, the extra mutual repulsion results in less energy being required to remove an electron.
- Hence less energy is required to remove an electron from oxygen.

First ionization energy of Sulfur is less than Phosphorous [S < P]

Phosphorous [15]: $1s^2\ 2s^2 2p^6\ 3s^2 3p^3$ | ↑ | ↑ | ↑ |

Sulfur [16]: $1s^2\ 2s^2\ 2p^4 3s^2 3p^4$ | ↑↓ | ↑ | ↑ |

- Despite the increased nuclear charge the electron is easier to remove
- The 3p electron removed from sulfur is a paired electron, whereas the 3p electron from phosphorous is an unpaired electron.
- When electrons are paired, the extra mutual repulsion results in less energy being required to remove an electron.
- Hence less energy is required to remove an electron from sulfur.

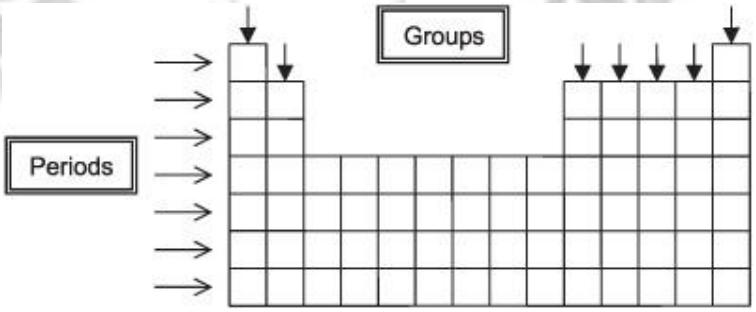
CHECKPOINT 4

Do the check point questions on page 55 from text book.

2B – 1 THE PERIODIC TABLE

SPECIFICATION OBJECTIVE 2.15 & 2.16

- In the modern periodic table the elements are arranged in order of their **atomic number**.
- Vertical columns in the periodic table are called **groups** and the horizontal rows are called **periods**.



  - All the elements within the **period** have the **same number of electron shells**. Example: Elements in period 2 have 2 electron shells
  - All the elements within a **group** have the **same number of electrons in their outer shell** and they have similar properties.

	GROUP 1	GROUP 2	GROUP 3	GROUP 4
outer electronic configuration	ns^1	ns^2	ns^2np^1	ns^2np^2

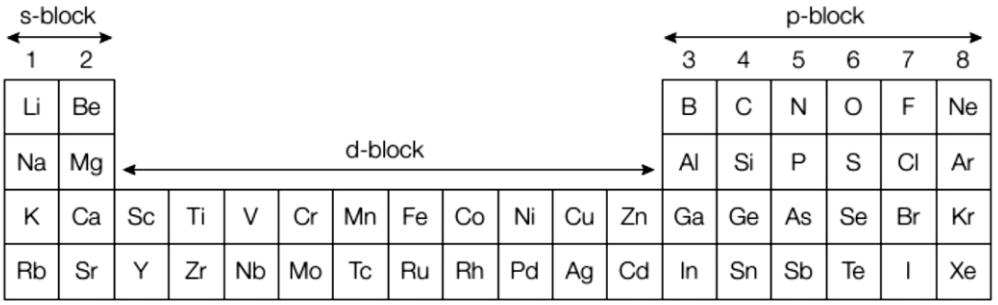
	GROUP 5	GROUP 6	GROUP 7	GROUP 8
outer electronic configuration	ns^2np^3	ns^2np^4	ns^2np^5	ns^2np^6

- Position of an element in the periodic table is determined by its electronic structure. The physical and chemical properties depend on the electronic configuration.

THE BLOCKS OF THE PERIODIC TABLE

The periodic table can be split into four blocks, based on the orbital in which the last electron gets filled.

- s – Block: elements of group 1 and 2 .Their outermost electrons are found in the s sub-shell.
- p – Block: elements in groups 3 to 8. Outermost electrons are found in p sub-shell.
- d – Block: transition elements and are found between group 2 and 3. Outermost electrons are found in d sub-shell.
- f – Block: lanthanides and actinides, and they have outermost electrons in the f sub-shell.



CHECKPOINT 5

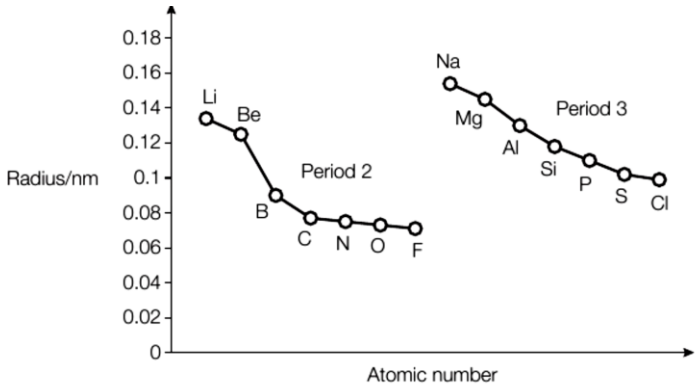
Do the check point questions on page 56 from text book.

2B – 2 THE PERIODIC PROPERTIES

SPECIFICATION OBJECTIVE 2.17, 2.18 part (i) (ii) (iii)

ATOMIC RADIUS

Atomic radius of an element is the measurement of the size of its atom. It is the distance from the center of the nucleus to the boundary of electron cloud. Since atoms does not have a well defined boundary, we can find the atomic radius by determining the distance between the two nuclei and dividing by 2.



The atomic radius generally decreases across a period

- Nuclear charge (proton number) increases across the period
- Shielding effect remains constant as the electrons are added to the same shell.
- Therefore, attraction between the nucleus and the electrons increases.
- Electrons are pulled closer to nucleus.

The atomic radius generally increases down the group.

- Electrons are added to new energy levels down the group
- Shielding effect increases
- The attraction between the electrons and the nucleus decreases.

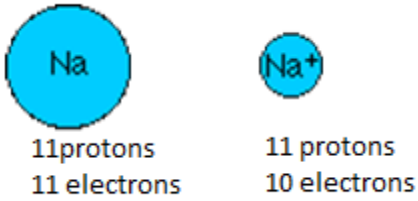
Shielding effect

- The electron in the outermost energy level experiences more inter-electron repulsion known as shielding.

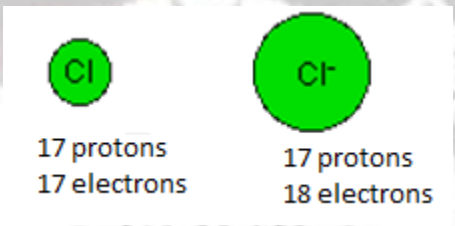
- The greater the number of the inner shell electrons the greater is the shielding effect (screening effect).

IONIC RADIUS

The radius of a positive ion is smaller than the element's atomic radius. This is because as the electrons are removed the remaining electrons are more strongly attracted to the positive nucleus.



The radius of a negative ion is larger than the element's atomic radius. This is because as the electrons are gained, the number of electrons increases and therefore are bound less tightly to the nucleus.



MELTING AND BOILING TEMPERATURES

Elements – As you move from left to right the elements go from highly electropositive metals through metalloids with giant structures to the simple molecular structure of non-metals.

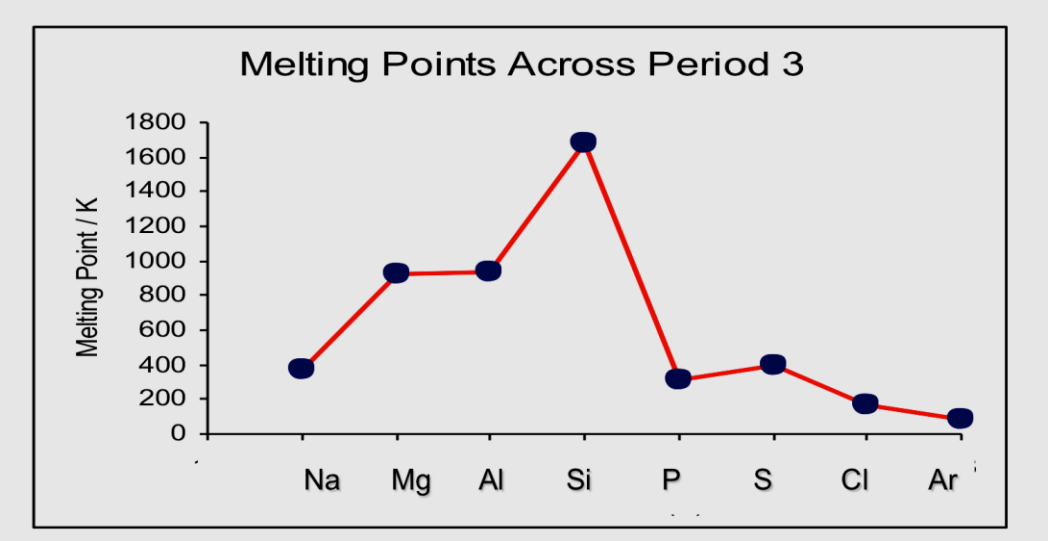
Both periods 2 and 3 show similar trends in their melting points and boiling points, which are linked to changes in structure and bond strength of these elements.

The table below illustrates the changes in melting and boiling points of elements in Periods 2 and 3

PERIOD 2	Li	Be	B	C (DIAMOND)	N	O	F
melting temperature / °C	181	1278	2300	3550	−210	−218	−220
boiling temperature / °C	1342	2970	3927	4827	−196	−183	−188
type of bonding	metallic	metallic	covalent	covalent	covalent	covalent	covalent
structure	giant lattice	giant lattice	giant lattice	giant lattice	simple molecular	simple molecular	simple molecular

PERIOD 3	Na	Mg	Al	Si	P	S	Cl
melting temperature / °C	98	649	660	1440	44	113	−101
boiling temperature / °C	883	1107	2467	2355	280	445	−35
type of bonding	metallic	metallic	metallic	covalent	covalent	covalent	covalent
structure	giant lattice	giant lattice	giant lattice	giant lattice	simple molecular	simple molecular	simple molecular

The graph below shows the melting points of the elements in Period 3



Element(s)	Structure and type of bonding	Trend in melting point
Metals (Eg: Na , Mg, Al)	Metallic bonding Giant lattice structure	Melting point increases across the period • Ionic size decreases across the period • More number of electrons in the 'sea' of electrons Therefore, strength of metallic bonding increases.
Macromolecules (Eg: Si)	Covalent bonding Giant covalent structure	Melting point is very high • They have strong covalent bonds linking all their atoms together A lot of energy is needed to break the strong covalent bonds
Simple molecules (Eg: P_4, S_8, Cl_2)	Covalent bonding Simple molecular structure	Melting point is low • Melting point depends on the strength of intermolecular forces (London forces) between molecules. • Since London forces are weak they have a low melting point. • London forces are stronger in S_8 as more electrons are present. P_4 contains more electrons than Cl_2.so London forces stronger in P_4 than Cl_2
Noble Gases (Eg: Ar)	Simple atomic structure	They have the lowest melting and boiling points because they exist as individual atoms resulting in very weak London forces between atoms

CHECKPOINT 6

Do the check point questions on page 59 from text book.